

Electrical Circuits I

ENGR 2405

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Chapter 08 – Second Order Circuits

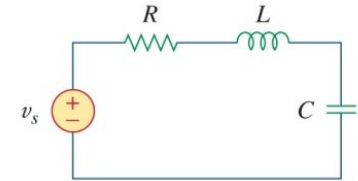
Austin Community College

Overview

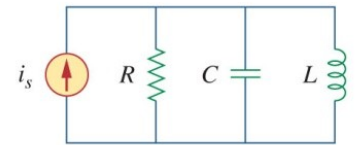
- This chapter will introduce second order circuits - those that need a second order differential equation
- RLC series and parallel circuits will be discussed in this context
- The step response of these circuits will be covered as well

Second Order Circuits

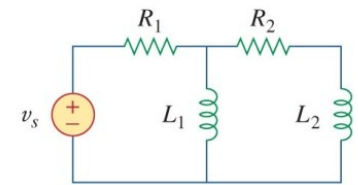
- When more than one “storage element”, that is, capacitor or inductor is present, the equations require second order differential equations
- The analysis is similar to what was done with first order circuits
- We will only consider DC independent sources - dependent second order circuits are an advanced topic



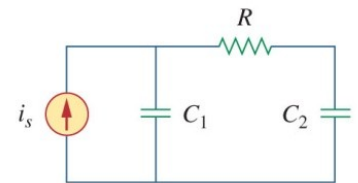
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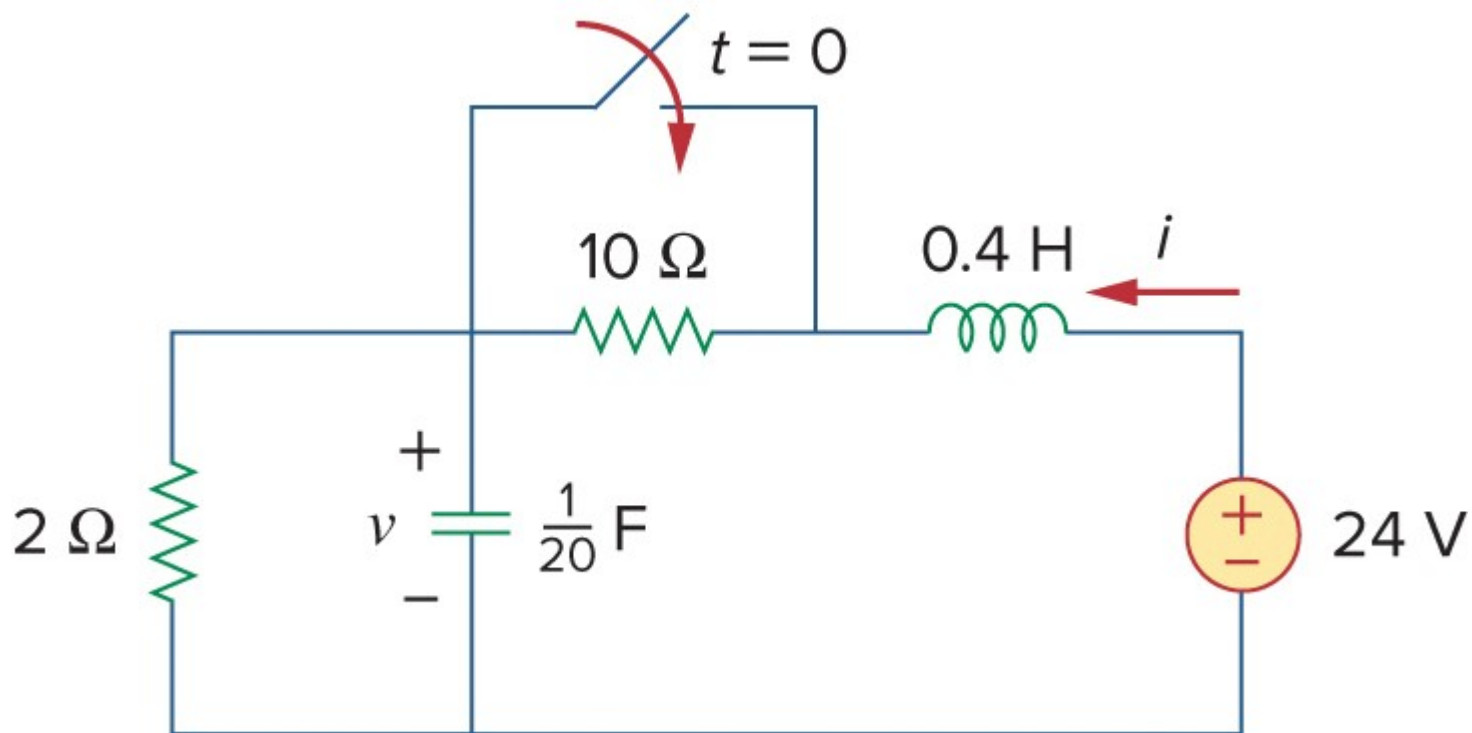
(d)

Finding Initial and Final Values

- Working on second order system is harder than first order in terms of finding initial and final conditions
- You need to know the derivatives, dv/dt and di/dt as well
- Getting the polarity across a capacitor and the direction of current through an inductor is critical
- Reminder: capacitor voltage and inductor current are always continuous

Practice Problems

8.1 Find $i(0+)$, $v(0+)$, $i'(0+)$, $v'(0+)$, $i(\infty)$, and $v(\infty)$

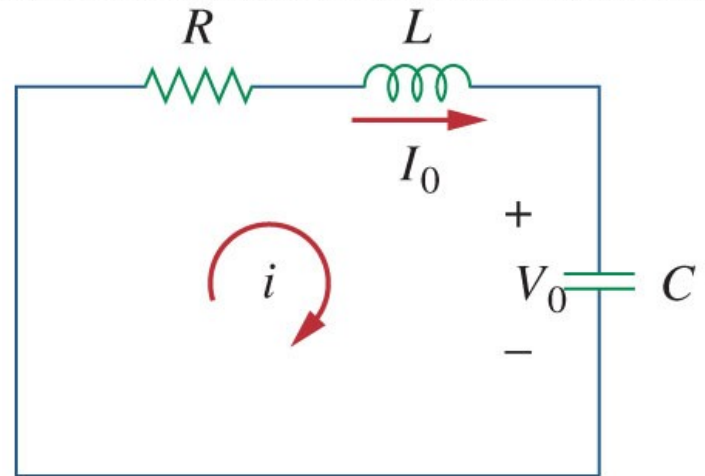


Source Free Series RLC

- The energy at $t = 0$ is stored in the capacitor, represented by V_0 and in the inductor, represented by I_0

$$v(0) = \frac{1}{C} \int_{-\infty}^0 i dt = V_0$$

$$i(0) = I_0$$



Source Free Series RLC

- Applying KVL around the loop:

$$Ri + L \frac{di}{dt} + \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau = 0$$

- The integral can be eliminated by differentiation:

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{i}{LC} = 0$$

- Here you can see the second order equation that results

Source Free Series RLC

- Two initial conditions are needed for solving this problem
- The initial current is given
- The first derivative of the current can also be had:

$$Ri(0) + L \frac{di(0)}{dt} + V_0 = 0$$

- Or

$$\frac{di(0)}{dt} = -\frac{1}{L} (RI_0 + V_0)$$

Source Free Series RLC

- Based on the first order solutions, we can expect that the solution will be in exponential form
- The equation will then be:

$$Ae^{st} \left(s^2 + \frac{R}{L}s + \frac{1}{LC} \right) = 0$$

- For which the solutions are:

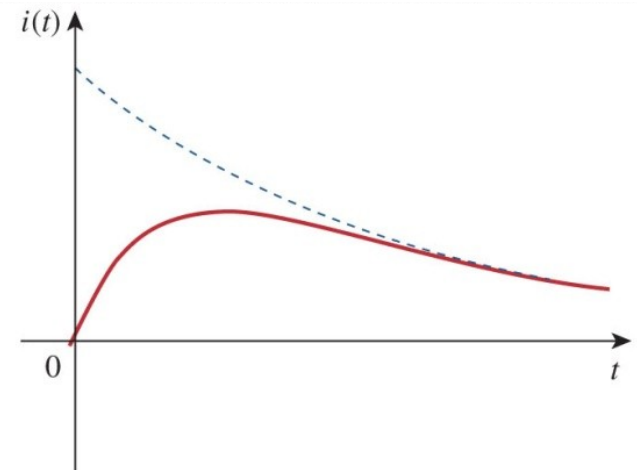
$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$
$$\alpha = \frac{R}{2L} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Overdamped ($\alpha > \omega_0$)

- When $\alpha > \omega_0$ the system is overdamped
- In this case, both s_1 and s_2 are real and negative
- The response of the system is:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

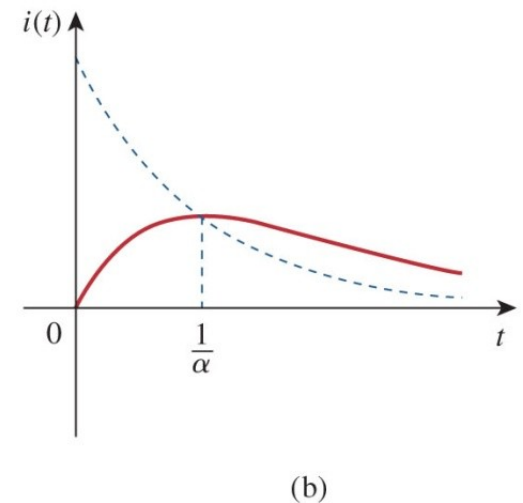
- From this, we should not expect to see an oscillation



(a)

Critically Damped ($\alpha = \omega_0$)

- When $\alpha = \omega_0$ the system is critically damped
- The solution to the DE become:
$$i(t) = (A_1 + A_2 t) e^{-\alpha t}$$
- There are two components to the response: an exponential decay and an an exponential decay multiplied by a linear term



Underdamped ($\alpha < \omega_0$)

- When $\alpha < \omega_0$ the system is critically damped
- The solution to the DE become:

$$i(t) = e^{-\alpha t} \left[A_1 e^{-j\omega_d t} + A_2 e^{-j\omega_d t} \right]$$

- Where:

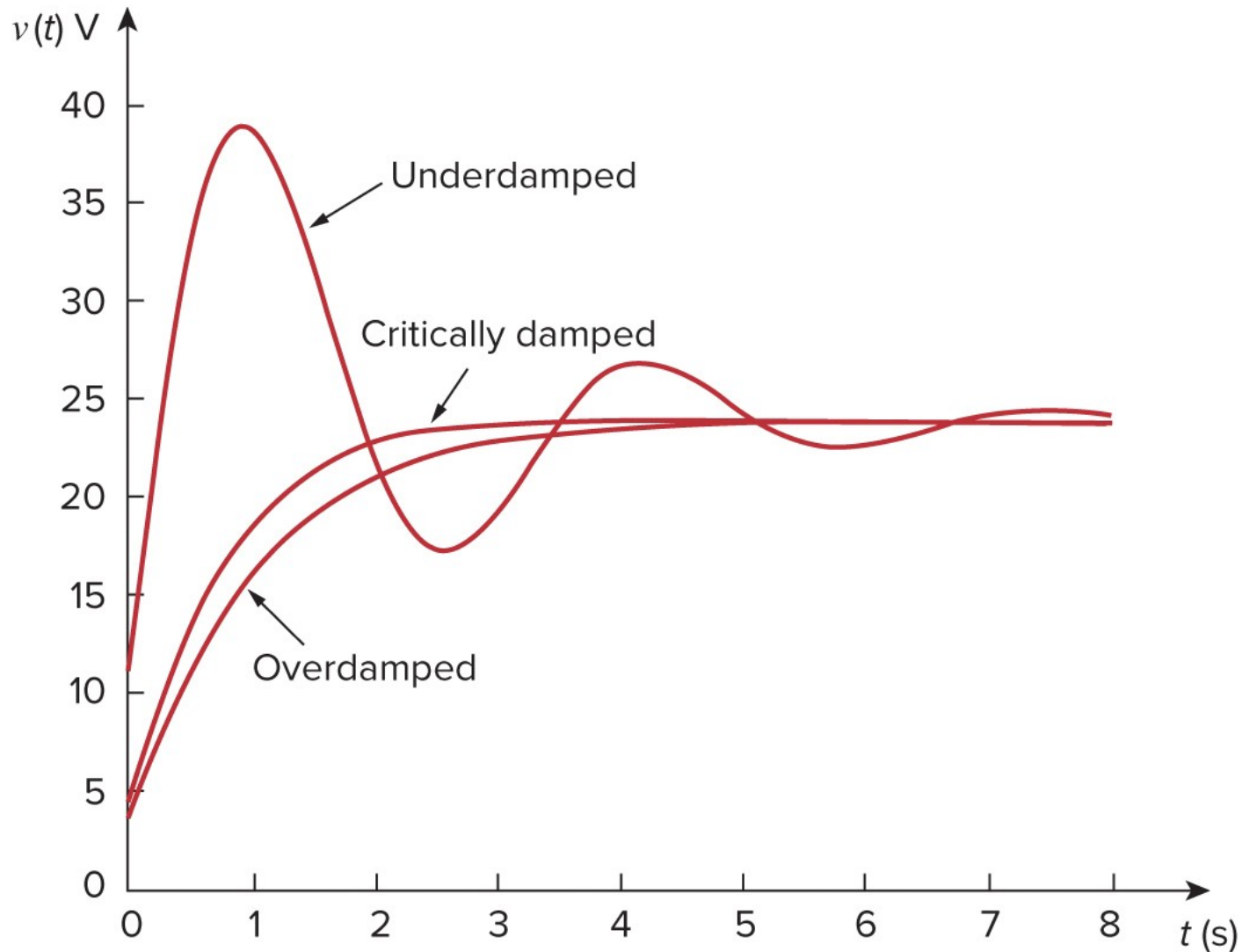
$$j = \sqrt{-1} \text{ and } \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

- Rearranging using Euler's formula and algebra:

$$i(t) = e^{-\alpha t} \left[B_1 \cos(\omega_d t) + B_2 \sin(-j\omega_d t) \right]$$

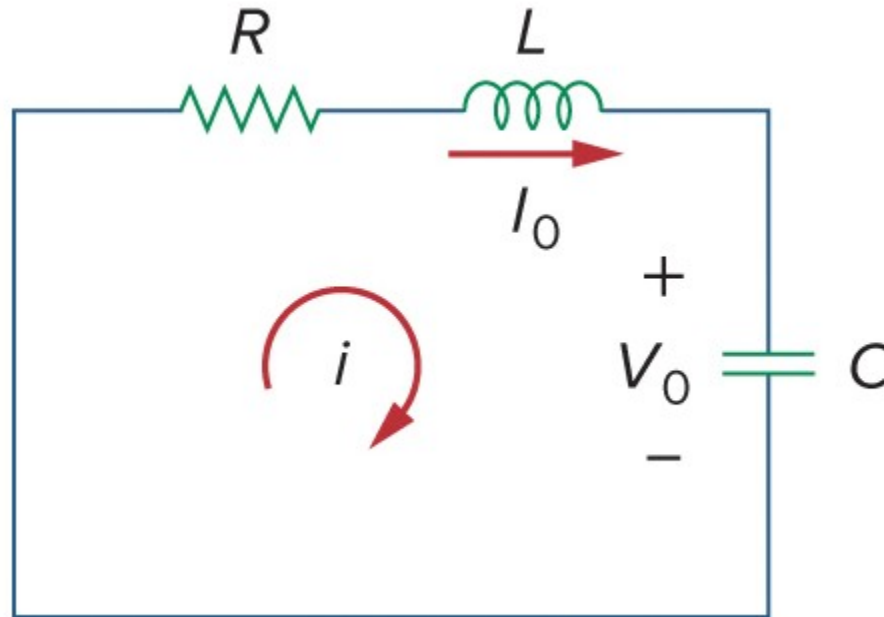
- ω_0 is called the undamped natural frequency
- ω_d is called the damped natural frequency

Damping Comparison



Practice Problems

8.3 If $R=10\Omega$, $L=5\text{H}$, and $C=2\text{mF}$, find α , ω_0 , s_1 , and s_2 . What is the natural response of the circuit?



Damping and RLC networks

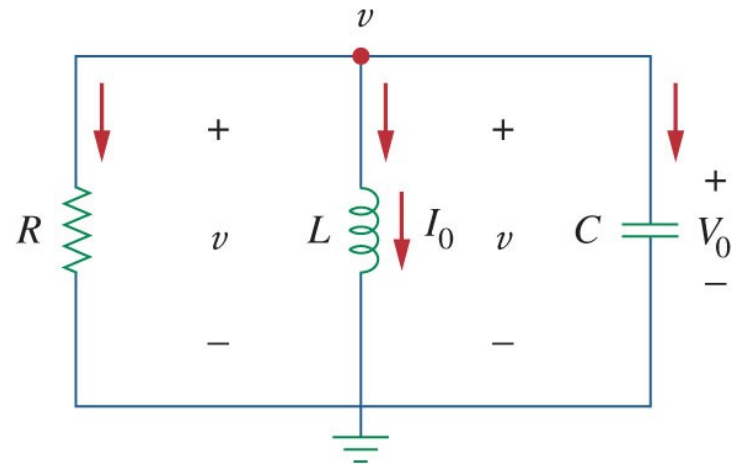
- RLC networks can be characterized by the following:
 1. The behavior of these networks is captured by the idea of damping
 2. Oscillatory response is possible due to the presence of two types of energy storage elements
 3. It is typically difficult to tell the difference between damped and critically damped responses

Source Free Parallel RLC Network

- Now let us look at parallel forms of RLC networks
- Assume the initial current and voltage to be:

$$i(0) = I_0 = \frac{1}{L} \int_{-\infty}^0 v(t) dt$$

$$v(0) = V_0$$



Source Free Parallel RLC Network

- Applying KCL to the top node we get:

$$\frac{v}{R} + \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau + C \frac{dv}{dt} = 0$$

- Taking the derivative with respect to t gives:

$$\frac{d^2 v}{dt^2} + \frac{1}{RC} \frac{dv}{dt} + \frac{1}{LC} v = 0$$

- The characteristic equation for this is:

$$s^2 + \frac{1}{RC} s + \frac{1}{LC} = 0$$

Source Free Parallel RLC Network

- From this, we can find the roots of the characteristic equation to be:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$$
$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

- As in last time, there are three scenarios to consider

Damping

- For the overdamped case, the roots are real and negative, so the response is:

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

- For critically damped, the roots are real and equal, so the response is:

$$v(t) = (A_1 + A_2 t) e^{-\alpha t}$$

- In the underdamped case, the roots are complex and so the response will be:

$$v(t) = e^{-\alpha t} \left[A_1 e^{-j\omega_d t} + A_2 e^{-j\omega_d t} \right]$$

Damping

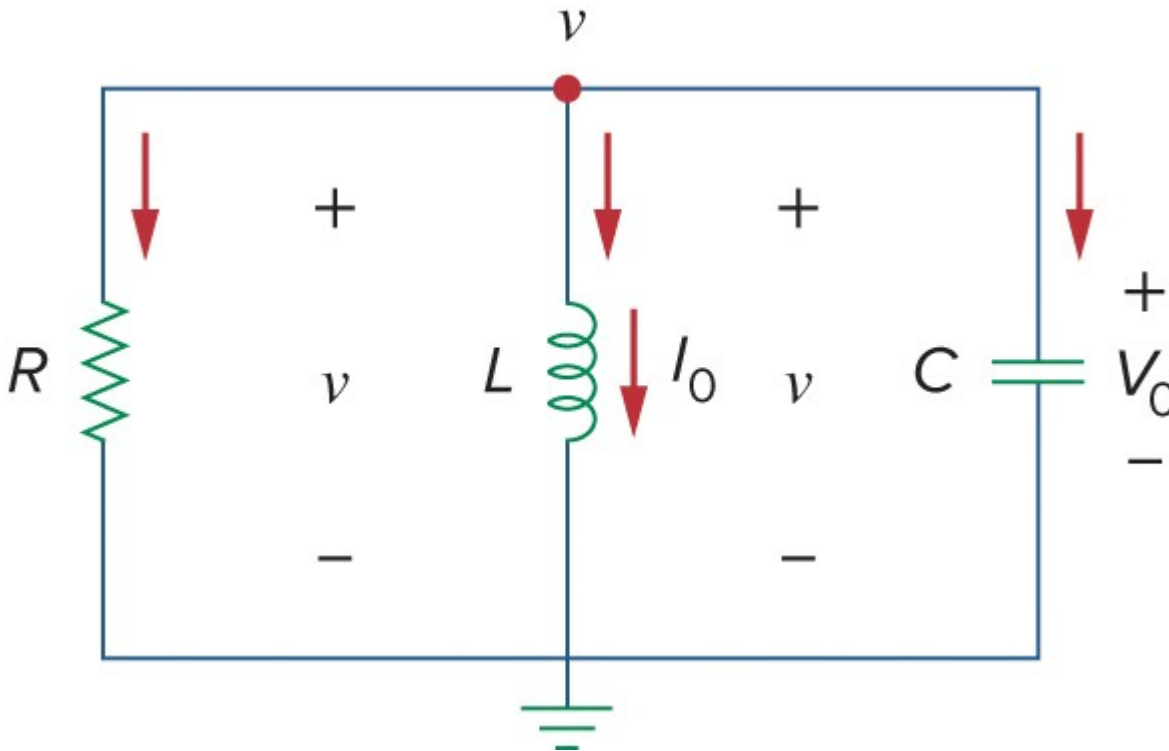
- To get the values for the constants, we need to know $v(0)$ and $dv(0)/dt$
- To find the second term, we can use:

$$\frac{V_0}{R} + I_0 + C \frac{dv(0)}{dt} = 0 \Rightarrow \frac{dv(0)}{dt} = -\frac{(V_0 + RI_0)}{RC}$$

- The voltage waveforms will be similar to those shown for the series network
- Note: In the series circuit, we first found the I_L and then solved for the rest from that while here we start with the V_C and solve for the other variables from it

Practice Problems

8.5 If $R=2\Omega$, $L=0.4\text{H}$, and $C=25\text{mF}$, $v(0)=0$, and $i(0)=50\text{mA}$, find $v(t)$, for $t>0$



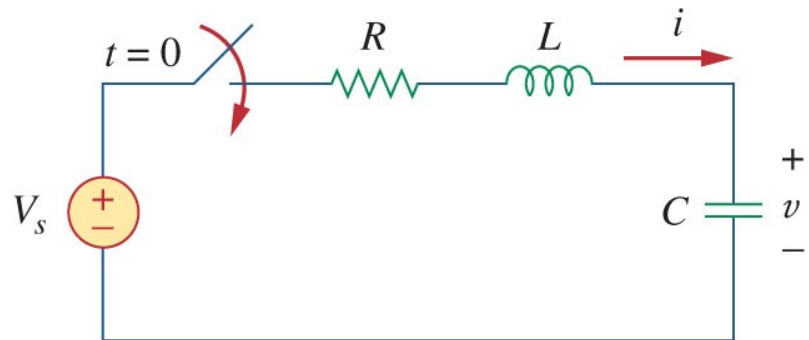
Step Response of a Series RLC Circuit

- Now let us consider what happens when a DC voltage is suddenly applied to a second order circuit
- Consider the circuit shown - the switch closes at $t = 0$
- Applying KVL around the loop for $t > 0$:

$$L \frac{di}{dt} + Ri + v = V_s$$

- but

$$i = C \frac{dv}{dt}$$



Step Response of a Series RLC Circuit

- Substituting for i gives:

$$\frac{d^2 v}{dt^2} + \frac{R}{L} \frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}$$

- This is similar to the response for the source free version of the series circuit, except the variable is different
- The solution to this equation is a combination of transient response and steady state

$$v(t) = v_t(t) + v_{ss}(t)$$

Step Response of a Series RLC Circuit

- The transient response is in the same form as the solutions for the source free version
- The steady state response is the final value of $v(t)$
 - in this case, the capacitor voltage will equal the source voltage

Step Response of a Series RLC Circuit

- The complete solutions for the three conditions of damping are:

$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Over-damped})$$

$$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Under-damped})$$

- The variables A_1 and A_2 are obtained from the initial conditions $v(0)$ and $dv(0)/dt$

Step Response of a Parallel RLC Circuit

- A similar treatment for the parallel RLC circuit reveals similar solutions
- The complete solutions for the three conditions of damping are:

$$i(t) = I_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad (\text{Over-damped})$$

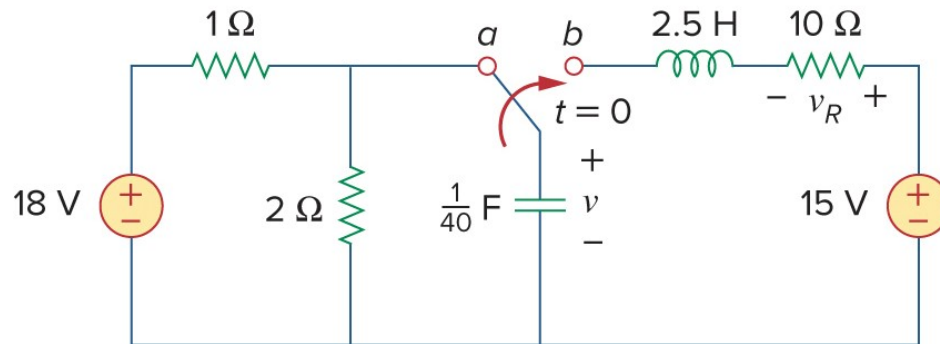
$$i(t) = I_s + (A_1 + A_2 t) e^{-\alpha t} \quad (\text{Critically damped})$$

$$i(t) = I_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad (\text{Under-damped})$$

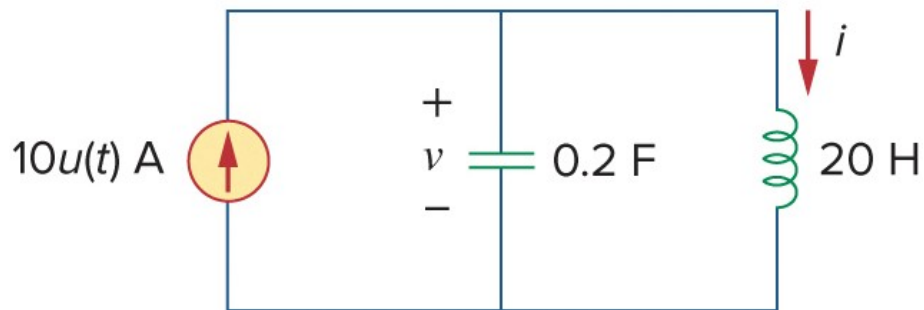
- Here, the variables A_1 and A_2 are obtained from the initial conditions $i(0)$ and $di(0)/dt$

Practice Problems

8.7 Find $v(t)$ and $v_R(t)$ for $t > 0$



8.8 Find $i(t)$ and $v(t)$ for $t > 0$



General Second Order Circuits

The principles of the approach to solving the series and parallel forms of RLC circuits can be applied to second order circuits in general:

The following four steps need to be taken:

1. First determine the initial conditions, $x(0)$ and $dx(0)/dt$
2. Turn off the independent sources and find the form of the transient response by applying KVL and KCL

Depending on the damping found, the unknown constants will be found

General Second Order Circuits

3. We obtain the steady state response as:

$$x_{ss}(t) = x(\infty)$$

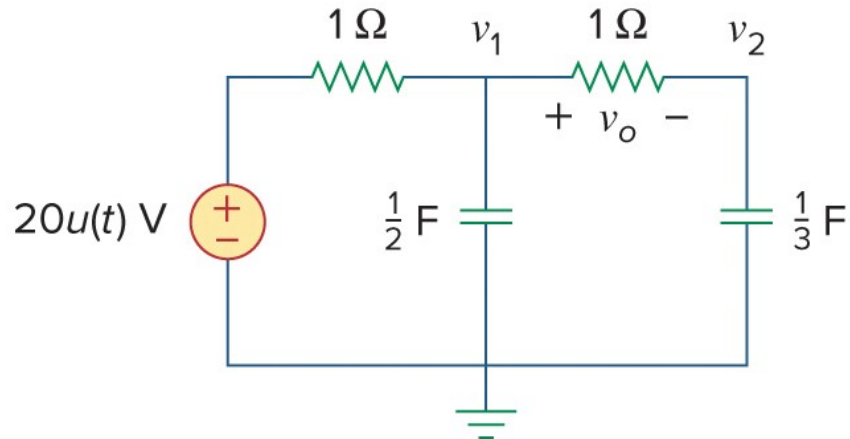
Where $x(\infty)$ is the final value of x obtained in step 1

4. The total response is now found as the sum of the transient response and steady-state response

$$x(t) = x_t(t) + x_{ss}(t)$$

Practice Problems

8.10 Find v and i for $t > 0$



8.11 If $v_s(t) = 10u(t)$, $R_1 = R_2 = 10\text{k}\Omega$, $C_1 = 20\mu\text{F}$, $C_2 = 100\mu\text{F}$, find $v_o(t)$

